

12.4 Assuming a general prior:

$$p(\mathbf{x}|\mathbf{z})p(\mathbf{z}) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I}) \mathcal{N}(\mathbf{z}|\mathbf{m}, \boldsymbol{\Sigma})$$

Using the result (2.115) to calculate the marginal probability:

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{W}\mathbf{m} + \boldsymbol{\mu}, \sigma^2\mathbf{I} + \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^T)$$

If we define $\boldsymbol{\mu}_0 = \mathbf{W}\mathbf{m} + \boldsymbol{\mu}$, we get the same mean as with a generalized prior.

$\boldsymbol{\Sigma}$ being a symmetric positive definite matrix, its Cholesky decomposition is $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$.

$$\implies \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^T = \mathbf{W}\mathbf{L}\mathbf{L}^T\mathbf{W}^T = (\mathbf{W}\mathbf{L})(\mathbf{W}\mathbf{L})^T$$

If we define $\mathbf{W}_0 = \mathbf{W}\mathbf{L}$, we get the same covariance as with a generalized prior.

$$\implies \mathcal{N}(\boldsymbol{\mu}_0, \sigma^2\mathbf{I} + \mathbf{W}_0\mathbf{W}_0^T) = \mathcal{N}(\mathbf{W}\mathbf{m} + \boldsymbol{\mu}, \sigma^2\mathbf{I} + \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^T)$$